

VIJAY WANKHEDE

Staff Backend Engineer · Fintech & Payments · Node.js / TypeScript · Kubernetes / AWS
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PROFESSIONAL SUMMARY

Staff-level backend engineer with 6+ years designing, scaling, and operating high-availability distributed systems for fintech, payments, and consumer-facing products. Deep Node.js / TypeScript expertise, comfortable with functional programming patterns (fp-ts, Either/Task, ADTs) and event-driven microservices on Kubernetes and AWS. Trusted technical partner across product, design, and engineering: lead architectural design reviews, raise the technical bar through mentorship and code review, and ship reliable, secure, performant services that hold up under real production load. Experienced building payments, ledger, and consumer-facing flows end-to-end, and integrating AI-powered tooling (Claude, Cursor) into the developer experience to accelerate delivery.

CORE COMPETENCIES

Staff Backend Engineering · High-Scale Distributed Systems · Payments & Ledger Systems · Consumer-Facing Products · Microservices Architecture · Event-Driven Architecture · API Design (REST, GraphQL, gRPC) · System Design & Architecture · Reliability, Security & Performance · Functional Programming (fp-ts) · Mentorship & Technical Leadership · Cross-Functional Collaboration · Stakeholder Alignment · Autonomy in Ambiguity

TECHNICAL SKILLS

Languages: TypeScript, Python, Go

Backend Frameworks: NestJS, Express, fp-ts (functional TypeScript), FastAPI, Flask, gRPC

Architecture: Microservices, Event-Driven Architecture, CQRS, Domain-Driven Design, Hexagonal Architecture, REST, GraphQL, gRPC, WebSockets

Messaging / Streaming: Apache Kafka, RabbitMQ, AWS SQS / SNS / EventBridge, Redis Streams

Databases: PostgreSQL, MySQL, DynamoDB, MongoDB, CouchDB, Redis (cache, pub/sub, locks)

System Design: Horizontal scaling, sharding, replication, caching strategies, idempotency, exactly-once processing, rate limiting, circuit breakers, retries with backoff, load balancing

Testing & Quality: Jest, Mocha, Pytest, integration tests, contract tests, property-based tests, load testing (k6, Artillery)

AI Tooling: Claude Code, Cursor, custom Claude skills for codegen, code review, and adversarial testing

PROFESSIONAL EXPERIENCE

Catalysis Network — Staff Backend Engineer / Tech Lead *Remote · 2025 – Present*

- Serve as the senior technical partner across product and engineering, leading design and implementation of high-scale backend systems from zero — service topology, API contracts, data models, and rollout strategy — for a platform launching with \$150M+ in user assets.
- Architected event-driven Node.js / TypeScript microservices on Kubernetes (AWS EKS) using NestJS and functional patterns (fp-ts-style Either/Task error handling), backed by Kafka, PostgreSQL, and Redis for sub-second consumer-facing read paths under sustained load.
- Owned the Router middleware end-to-end: load-balanced services behind an API gateway, idempotent write paths, optimistic concurrency, and replay-safe Kafka consumers — guaranteeing exactly-once semantics for restake, reward, and slashing flows.
- Designed and shipped a signature-based settlement flow using EIP-2771 meta-transactions (off-chain signer, validation pipeline, on-chain adapter), eliminating user friction in the consumer-facing path

while preserving full cryptographic auditability.

- Stood up the reliability and observability stack from scratch — Prometheus metrics, Grafana dashboards, structured JSON logs, Sentry alerting, SLOs for critical write paths (99.9%) — and authored the on-call runbooks the team uses today.
- Drove engineering velocity by integrating Claude Code and Cursor into the developer experience: built custom Claude skills for adversarial test generation, architectural review, and codegen, cutting feature lead time on the StakeManager service by an estimated 40%.
- Mentored engineers, ran weekly design reviews, and raised the technical bar through structured PR review and shared TypeScript / functional-programming standards.

Synthr Protocol — Senior Backend Engineer *Remote · 2023 – 2025*

- Led the full backend stack for a production cross-chain transfer / payments product across Axelar and LayerZero, owning the relayer service, event indexer, off-chain verification layer, and REST / GraphQL APIs powering consumer-facing flows and partner integrations.
- Re-architected the message pipeline around Apache Kafka with consumer groups, dead-letter queues, and exactly-once semantics, lifting customer retention by 30% by eliminating stuck transactions and exposing real-time status to end users.
- Designed and shipped a TypeScript / NestJS event indexer ingesting logs from 6+ chains into PostgreSQL, with reorg handling, backfill, and a Redis-backed cursor store for resumable consumption — production-grade reliability under bursty load.
- Stood up Grafana + Prometheus monitoring from scratch, defined SLIs/SLOs for relay latency and indexer lag, and built alerting that paged on-call before users noticed degradation, materially improving platform reliability.
- Hardened the off-chain verification layer against timing exploits and data integrity issues with deterministic replay tests and invariant checks wired into CI; led the security review process for every backend change.
- Set TypeScript coding standards, reviewed every backend PR, and onboarded three engineers across the relayer and indexer codebases.

Instabit — Backend Engineer (Payments & Fintech) *Remote · 2021 – 2023*

- Built a production payments confirmation pipeline in Go and Python handling thousands of transaction events per minute, with idempotent processing, mempool monitoring, and configurable confirmation thresholds per merchant.
- Designed the merchant-facing REST API and webhook delivery system in FastAPI, including HMAC-signed payloads, exponential-backoff retries, and a dead-letter queue for poison messages — patterns I now treat as table stakes for any payments integration.
- Migrated the core ledger from a single-node PostgreSQL instance to a replicated, partitioned setup with read replicas, cutting p99 read latency by ~60% and unblocking the analytics team to ship customer-facing reporting.
- Containerized services with Docker, deployed to AWS, set up GitHub Actions CI/CD with automated migrations, and introduced structured logging that became the basis for the team's incident response process.

Deqode Labs — Software Engineer *Pune, India · 2019 – 2021*

- Built TypeScript / Node.js backend services on NestJS for fintech and supply-chain clients, including authentication (JWT + refresh rotation), role-based access control, audit logging, and high-volume event listeners ingesting on-chain data into PostgreSQL.
- Wrote integration test suites with Jest and Supertest, raising backend coverage on the flagship project from ~35% to ~80% and catching a class of race conditions before production.
- Collaborated directly with clients on API design, system trade-offs, and deployment planning — early experience that shaped how I run technical design reviews and stakeholder alignment today.

EDUCATION

B.E., Computer Engineering — Institute of Engineering and Technology, India